

Vishal Singh

Full Stack Engineer

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SUMMARY

Full Stack Engineer with 2+ year of experience building and shipping production web applications using React, Next.js, Node.js, and PostgreSQL. Skilled in microservices architecture, REST API design, CI/CD pipelines, and cloud deployments on AWS and Vercel. Delivered 5+ projects in Agile/Scrum environments with measurable performance outcomes.

WORK EXPERIENCE

Full Stack Developer

Jan 2025 – Present

Freelancer | Remote

- Shipped 5+ full-stack products (React, Next.js, Node.js, PostgreSQL) across SaaS and education domains, achieving 95+ Lighthouse scores on all deliverables.
- Reduced deployment time from hours to under 5 minutes by implementing GitHub Actions CI/CD pipelines with Docker and AWS/Vercel cloud infrastructure.

Backend Developer Intern

Jun 2024 – Nov 2024

Better Auth | Remote

- Built secure REST APIs (Node.js, TypeScript) with JWT, OAuth 2.0, and rate limiting; improved PostgreSQL query execution time by 40% through indexing and ORM optimisation.

PROJECTS

RGD Academy — SaaS School Management Platform

GitHub | Live

Next.js 15 • React 18 • TypeScript • PostgreSQL • Prisma ORM • NextAuth.js • Tailwind CSS • Razorpay • Upstash Redis • GitHub Actions

- Built 3 role-based portals (Admin, Staff, Student) serving 200+ users with Next.js SSR/SSG, middleware route protection, and Aadhar-based multi-provider auth.
- Integrated Razorpay payments with auto-generated PDF receipts processing 500+ student fee records; built exam module with marksheet generation and grade/rank computation.
- Automated student promotions, attendance alerts, and fee reminders via Vercel Cron Jobs, saving 10+ admin hours per week with 99.9% uptime over 3 months.

Astra — AI-Native Engineering Platform

GitHub

TypeScript • Next.js 15 • React 19 • Node.js • PostgreSQL (pgvector) • Redis • Docker • Kubernetes • LangGraph • Socket.io • Prometheus/Grafana

- Built a 7-agent autonomous AI system with LangGraph that generates and deploys software from natural language requirements, cutting prototyping time by 60%.
- Shipped real-time collaboration via Socket.io supporting 50+ concurrent users with presence tracking and cursor sync; achieved P95 API latency under 200ms with Prometheus/Grafana observability.

NexPay — Multi-Tenant Payment Infrastructure

GitHub

Node.js • TypeScript • Next.js 14 • PostgreSQL • Redis • BullMQ • Docker • NGINX • Stripe/Razorpay • Turborepo • Playwright

- Implemented double-entry ledger with PostgreSQL advisory locks and a fraud detection engine scoring payments 0–100 across 6 risk dimensions, achieving 92% precision.
- Wrote 100+ automated tests (unit, integration, E2E via Playwright) covering payment flows, refunds, webhook delivery, and tenant isolation with 94% code coverage.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, Python, Node.js, Bun
Frontend	React, Next.js (SSR/SSG/ISR), Tailwind CSS, Recoil, Framer Motion, Responsive Design
Backend / APIs	Express.js, REST APIs, GraphQL, Zod, OpenAPI, Serverless, Microservices, Auth & AuthZ
Databases	PostgreSQL, MongoDB, Prisma ORM, Indexing, Transactions, Query Optimisation, pgvector
Distributed	Redis, Kafka, Pub/Sub, WebSockets, gRPC, BullMQ, Caching, Horizontal Scaling
DevOps & Cloud	Docker, Kubernetes, GitHub Actions, NGINX, AWS (EC2, S3, Lambda), Vercel, Fly.io
Observability	Prometheus, Grafana, Structured Logging, JWT, OAuth 2.0, CORS/CSRF, Rate Limiting
Testing	Jest, Vitest, Playwright, Cypress, Supertest, Integration Testing, TDD

EDUCATION

B.Tech — Computer Science Engineering

KSVCEM, AKTU Lucknow

CGPA: 7.5 / 10

2022 – 2026